Dashboard Design & Visual Specification

The executive dashboard architecture translates the thirty analytical questions into a widescreen visual command center. The user interface enforces cognitive ergonomics, sub-second cross-filtering, and a strict one-visual-per-question standard.

Screen Layout & Viewport Architecture

- **16:9 Screen Fit & Zero Vertical Scroll:** The complete command center interface fits within a standard 1920x1080 display viewport without requiring vertical document scrolling, allowing executive decision-makers to evaluate performance within a single eye span.
- **Zone A: Sticky Global Navigation and Controls:** Occupying the top navigation strip across the full screen width, this zone houses the brand header, four primary stakeholder tabs (CEO, COO, CFO, Customer Churn), contextual decision sub-buttons for multi-view dashboards, three global interactive slicers (Country, Segment, Order Year), a slicer reset button, a live order coverage counter, and the interactive Decision Simulator trigger.
- **Contextual Decision Banner:** Positioned immediately below the global control bar, this full-width callout strip displays an uppercase view category pill, the strategic decision title, an executive context summary, and an accented primary decision rule container that changes dynamically with view selection.
- **Zone B: Five-Card Headline Metric Ribbon:** Arranged in a five-column grid across the screen width, this zone presents five live-calculated metric cards tailored to the active stakeholder role. Each card includes the headline metric value, a contextual benchmark subtitle, a target status badge, and a semantic left-border color accent.
- **Zone C: Primary Strategic Visual Row:** Structured as a balanced two-column grid (50% left, 50% right) with a fixed 255-pixel container height, this row visualizes primary strategic patterns, housing Question 1 and Question 2 through macro-level charts including spatial choropleth maps, line trends, and waterfall bridges.
- **Zone D: Diagnostic & Action Visual Row:** Structured as an equal three-column grid (33% left, 33% center, 33% right) with a fixed 255-pixel container height, this row visualizes root-cause diagnostics, distribution curves, and prioritized action tables, housing Question 3, Question 4, and Question 5.

Stakeholder Visual Encodings & Plot Mappings

CEO View Visual Mappings

- **Q1.1 Map Visual (Row 1 Left):** An interactive SVG World Choropleth Map rendering seven operating country polygons shaded by total net sales volume using light blue to dark navy gradations. Proportional glowing revenue bubbles indicate relative order density, supported by callout badges displaying country sales, order volume, and margin percentage. Clicking any country polygon cross-filters the entire dashboard to that territory.
- **Q1.2 Line Visual (Row 1 Right):** A multi-series line chart plotting transaction order year (2021 through 2025) along the horizontal axis and total net sales along the vertical axis, with five distinct lines representing the South, Central, West, East, and North operating regions.
- **Q1.3 Scatter Visual (Row 2 Left):** A four-quadrant scatter plot mapping average customer acquisition cost along the horizontal axis and average customer lifetime value along the vertical axis, plotting country nodes colored teal for high-equity markets and amber for lagging territories.
- **Q1.4 Bar Visual (Row 2 Center):** A 100% stacked horizontal bar chart displaying operating countries along the vertical axis and customer percentage share along the horizontal axis, segmented into color stacks for VIP, Premium, Business, and Consumer tiers.
- **Q1.5 Bar Visual (Row 2 Right):** A ranked horizontal bar chart plotting the top ten metropolitan cities sorted descending by total net profit, rendered with navy fill bars.

COO Button 1 Visual Mappings

- **Q1.1 Bar Visual (Row 1 Left):** A ranked horizontal bar chart plotting return reasons along the vertical axis and returned order volume along the horizontal axis, encoding controllable operational defects in coral red and buyer preference reasons in slate gray.
- **Q1.2 Heatmap Visual (Row 1 Right):** A matrix heatmap table crossing eight major suppliers along rows and six primary product categories along columns, conditionally formatting return rate percentages with coral red for high-defect cells above 8.5%, amber for warning cells between 6.0% and 8.4%, and emerald for compliant cells below 6.0%.

- **Q1.3 Bar Visual (Row 2 Left):** A clustered horizontal bar chart comparing the top seven merchandise subcategories along the vertical axis against incident counts, splitting bars into operational delay friction and catalog expectation mismatch.
- **Q1.4 Column Visual (Row 2 Center):** A vertical column chart plotting the top ten warehouse hubs along the horizontal axis and combined delayed and returned order volume along the vertical axis, highlighting defect-prone warehouses WH-004 and WH-005 in coral red.
- **Q1.5 Bar Visual (Row 2 Right):** A 100% stacked horizontal bar chart displaying product categories along the vertical axis and review sentiment proportions (Positive in teal, Neutral in amber, Negative in coral red) along the horizontal axis.

COO Button 2 Visual Mappings

- **Q2.1 Flow Visual (Row 1 Left):** A Sankey-style proportional flow diagram mapping order transitions from payment statuses (Paid, Pending, Failed, Refunded) into final order fulfillment statuses (Completed, Cancelled, Returned, Pending), displaying flow counts and completion percentages.
- **Q2.2 Column Visual (Row 1 Right):** A stacked vertical column chart plotting warehouse fulfillment centers along the horizontal axis and pending order backlog along the vertical axis, stacking series into urgent paid pending orders in amber, unpaid pending orders in blue, and cancelled orders in coral red.
- **Q2.3 Heatmap Visual (Row 2 Left):** A matrix heatmap table crossing four sales channels along rows and four payment methods along columns, conditionally formatting cell cancellation rates with red accents above 9.0%.
- **Q2.4 Distribution Visual (Row 2 Center):** A grouped percentile distribution chart plotting the four outbound shipping methods along the horizontal axis and shipping cost as a percentage of net sales along the vertical axis, contrasting average freight costs against 90th-percentile small-basket cost spikes.
- **Q2.5 Bar Visual (Row 2 Right):** A horizontal bar chart plotting the top product SKUs along the vertical axis and lost net revenue from cancelled and pending backorders along the horizontal axis, rendered in coral red.

CFO Button 1 Visual Mappings

- **Q1.1 Waterfall Visual (Row 1 Left):** A financial waterfall bridge chart stepping from gross sales through negative discounts, positive tax and shipping collections, realized net sales, negative product cost of goods sold, and final realized net profit, utilizing color-coded directional bars.
- **Q1.2 Line Visual (Row 1 Right):** A binned line chart plotting discount rate percentage bands along the horizontal axis and average realized profit margin along the vertical axis, bounded by a shaded margin area and an overlaid dashed 30% corporate margin floor.
- **Q1.3 Bar Visual (Row 2 Left):** A diverging horizontal bar chart plotting promotional coupon code tiers along the vertical axis and average profit margin along the horizontal axis, applying green formatting above 40%, amber between 30% and 39%, and coral red below the 30% margin threshold.
- **Q1.4 Column Visual (Row 2 Center):** A clustered vertical column chart plotting marketing campaign names along the horizontal axis and comparing net profit earned in teal against total discount allowance surrendered in coral red.
- **Q1.5 Bar Visual (Row 2 Right):** A grouped horizontal bar chart displaying customer segments along the vertical axis and average discount rate received along the horizontal axis, clustered by customer lifecycle status (Loyal, Returning, New).

CFO Button 2 Visual Mappings

- **Q2.1 Bar Visual (Row 1 Left):** A ranked horizontal bar chart plotting the ten marketing acquisition channels along the vertical axis and their CLV-to-CAC payback multiple along the horizontal axis, styling top-three channels in teal and bottom-three channels in coral red.
- **Q2.2 Area Visual (Row 1 Right):** A multi-series stacked area growth chart plotting customer purchase counts from initial order to fourteen or more orders along the horizontal axis and cumulative customer lifetime value along the vertical axis, stratified across customer segments.
- **Q2.3 Bullet Visual (Row 2 Left):** A clustered column chart displaying customer segments along the horizontal axis and contrasting average first-order net profit in teal against average acquisition cost in coral red.

- **Q2.4 Range Visual (Row 2 Center):** A dumbbell range chart displaying merchandise categories along rows, connecting realized net margin nodes in teal and catalog list markup nodes in navy through an amber margin gap bar.
- **Q2.5 Column Visual (Row 2 Right):** A 100% stacked column chart plotting operating transaction currencies along the horizontal axis and breaking down revenue share into net profit, product cost of goods sold, and tax and freight burdens.

Customer Churn View Visual Mappings

- **Q1.1 Heatmap Visual (Row 1 Left):** A cohort retention triangle heatmap matrix plotting customer acquisition cohorts along rows and purchase lifecycle stages (Order 1 through Order 16) along columns, conditionally formatting retained percentages to highlight attrition cliffs between Order #1 to #2 and Order #3 to #4.
- **Q1.2 Column Visual (Row 1 Right):** A stacked vertical column chart displaying customer segments along the horizontal axis and total customer lifetime value along the vertical axis, stacked into Active equity in teal, At-Risk equity in amber, and Churned equity in coral red.
- **Q1.3 Spline Visual (Row 2 Left):** A dual-spline line chart plotting delivery SLA delay gap days (-2 days early to +4 days late) along the horizontal axis and post-order churn probability along the vertical axis, contrasting standard deliveries in navy against defect-impacted orders in coral red.
- **Q1.4 Heatmap Visual (Row 2 Center):** A win-back sensitivity matrix table crossing promotional win-back offer tiers along rows and customer segments along columns, displaying projected net profit margins conditionally formatted to highlight the profitability of zero-discount point nudges over deep coupons.
- **Q1.5 Table Visual (Row 2 Right):** A prioritized CRM action queue table listing high-value At-Risk customer identifiers, segments, lifetime value equity with proportional inline teal data bars, unredeemed loyalty point balances, and preferred marketing channels, accompanied by an interactive CRM queue dispatch button.

Color Standards & Visual Hierarchy

- **Semantic Four-Color Palette:** The design strictly enforces unambiguous color meanings across all thirty charts: primary navy (#1E3A8A) and blue (#2563EB) represent volume, demand, and gross revenue scale; teal (#0D9488) and emerald (#10B981) denote bottom-line net profit, high retention, and healthy margins; amber (#D97706) highlights pending fulfillment queues, moderate discount bands, and at-risk accounts; and coral red (#E11D48) signals customer churn, delivery delays, operational returns, and margin-eroding promotions.
- **Visual Typography & Ratio Contrast:** Metric cards and visualization headers maintain prominent typography where aggregate numerical figures appear at twice the scale of supporting benchmark text, allowing instant scanning of operational status.
- **Consistent Container Dimensions:** Every visual container adheres to a standardized height of 255 pixels, eliminating jagged alignment and maintaining horizontal grid harmony across both two-column and three-column visual rows.

Interactive Behaviors & Control Mechanisms

- **Multi-Tier Navigation Routing:** Selecting any top-level stakeholder tab instantly refreshes the active view, the headline KPI ribbon, and the five analytical visual containers without triggering page reloads. Selecting contextual sub-buttons within the COO and CFO views alters the analytical focus between operational auditing and financial optimization.
- **Global Dimensional Slicing:** Adjusting the country, segment, or year dropdown selectors instantly filters the underlying dataset, re-aggregates headline KPI figures, and recalculates all five active chart canvases within 150 milliseconds.
- **Spatial Map Cross-Filtering:** Clicking any country polygon on the interactive SVG World Map applies an immediate territorial filter across the entire command center, updating metrics and charts to reflect only that country's orders. Clicking the active country again or selecting the reset control clears the filter back to the full enterprise dataset.
- **Executive Policy Simulator Modal:** Opening the policy simulator surfaces three dynamic sliders: maximum checkout discount cap (15% to 45%), operational return reduction rate (10% to 80%), and at-risk customer win-back rate (5% to 50%). Moving any slider recalculates price elasticity and retention equity at sixty frames per second,

updating projected dollar recovery for recovered discount margin, saved return revenue, and protected customer equity.

- **CRM Retention Action Dispatch:** Clicking the action trigger within the customer churn queue simulates exporting prioritized At-Risk accounts into an automated marketing workflow, validating account selections and generating an on-screen toast confirmation.